Assignmen1 (Question2)

G25AI1016; Name: Vijay Sri Krishna Devabhaktuni ①

Task 1 - Analytical Message Complexity.

Given:

N = total nodes

k = clusters

$n_c = N/k$ = nodes per cluster.

Tier1: Intra cluster (Ricart-Agrawala)

A node waiting to enter CS sends request to $(n_c - 1)$ other nodes in its cluster. Each of node sends back a reply.

Total Tier 1 messages $= 2 \times (n_c - 1)$

$= 2 \times (N/k - 1)$

{ substituting $n_c = N/k$ }

$= 2N/k - 2$

Tier2: Inter cluster (Suzuki - kasami token)

After getting all RA replies, the node becomes the Cluster Gateway. Gateway sends token request to $(k-1)$ other cluster gateway = $k-1$ messages.

Current token holder sends token back = 1 message

Total Tier2 messages = $(k-1)+1 = k$

Total $M(N,k) = \text{Tier 1} + \text{Tier 2}$

$$M(N,k) = (2N/k - 2) + k$$
$$= 2N/k + k - 2$$

$$\boxed{M(N,k) = 2N/k + k - 2}$$

Task 2 - Mathematical optimization

Minimize $M(N,k) = 2N/k + k - 2$

where N is a large constant and k is continous real variable, $k > 0$

Step 1 - first derivative

~~write as:~~ $\cancel{M = 2N \cdot k^{(-1)} + k - 2}$

$\cancel{dM/dk = -2N \cdot k^{(-2)}}$

write as $M = 2k^{(-1)}N + k - 2$

$$dM/dk = -2N \cdot (k)^{-2} + 1$$
$$= 2N/k^2 + 1$$

Step 2 - set first derivative to zero

$$-2N/k^2 + 1 = 0$$
$$1 = 2N/k^2.$$
$$k^2 = 2N$$

$\boxed{k^* = \sqrt{2N}}$ ← optimal number of clusters

Step 3 - Second derivative test (to confirm this is a minimum).

$$(d)^{2M} / d(k)^2 = d/dk \left[-2N \cdot (k)^{-2} + 1 \right]$$

$$= (-2) \times (-2N) \cdot \times (k)^{-3}$$

$$= 4N / (k)^3$$

Since $N > 0$ and $k > 0$ we have $4N/k^3 > 0$ for all valid k therefore $\frac{(d)^{2M}}{d(k)^2} > 0$ which means k^* is a minimum (proved)

Step 4 - Substitute k^* to find minimum message Complexity.

$$M(N, k^*) = 2N / (\sqrt{2N}) + (\sqrt{2N}) - 2$$

$$\frac{2N}{\sqrt{2N}} = \sqrt{2N} \qquad (\because \sqrt{2N} \times \sqrt{2N} = 2N)$$

$$M(N, k^*) = \sqrt{2N} + \sqrt{2N} - 2$$

$$= 2 * \sqrt{2N} - 2$$

$$M_{min} = 2\sqrt{2N} - 2$$

Asymptotic Complexity = $O*(\sqrt{N})$

4

Task 3 (a) – Graph of $M(72, k)$ vs k for

$N = 72.$

Subtitude $N = 72$:

$$M(72,k) = 144/k + k - 2$$

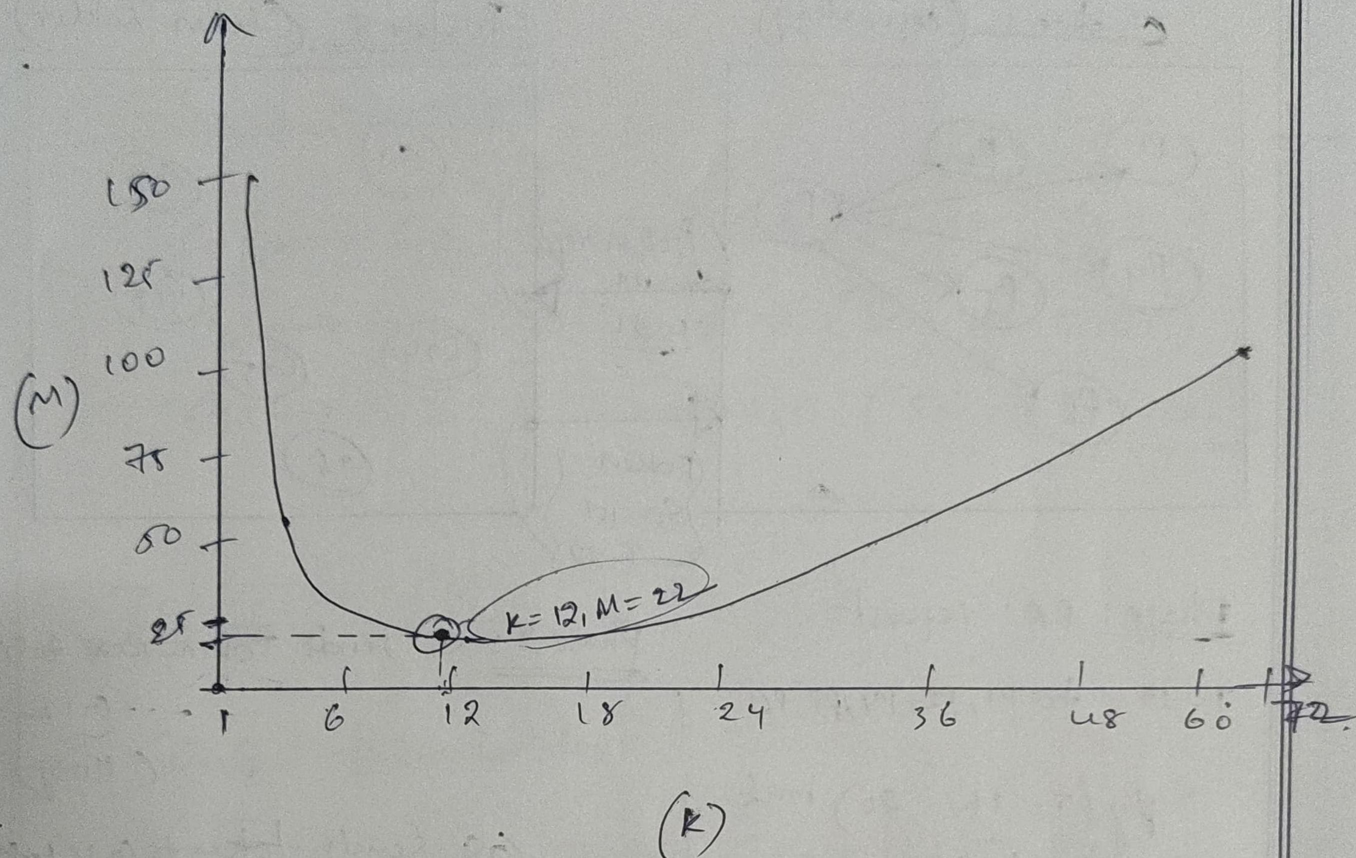
$$k^* = \sqrt{2 \times 72} = \sqrt{144} = 12$$

$$nc = 72/12 = 6$$

*minimum

k	M(72,k)
1	143
2	72
3	49
4	38
6	28
8	24
10	22.4
.	

k	M(72,k)
12	22
14	22.9
18	24
24	28
36	38
72	72..



$$\boxed{M(72, K) = 144/K + K - 2}$$

Task 3(b) - Message - flow graph (N = 72, K = 12, nc = 6)

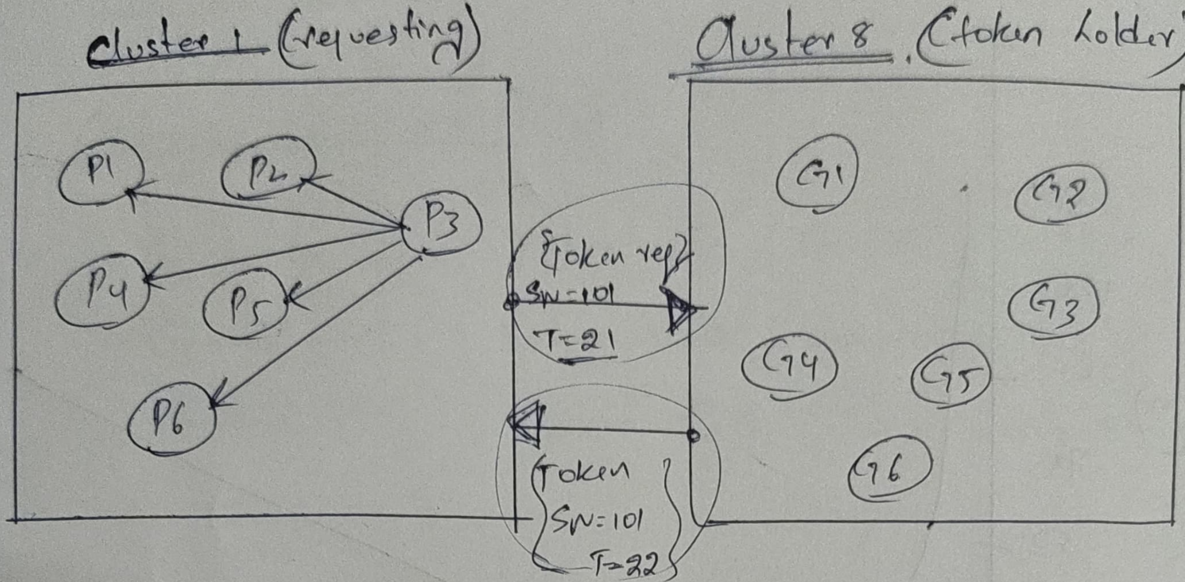
Configuration : K = 12 clusters nc = 6 nodes each.

Assume : Requesting node = P3

Gateway = G1 | Token holder = G8

Initial Lamport timestamp = 15.

Sequence number = 101



Phase 1 RA: request

T= 15 to P1, P2, P4, P5, P6

Reply (T= 16...20) back to P3

Phase 2: G1 sends Token-ReQ to G2 ... G12 (11 msg)

G8 sends token to G1 (1 msg)

P3 recieves token ---→ Enters Critical Section

Lamport clock Rules:

On Send: Local clock = local clock +1 (attach to message)

on recieve: Local clock = max(local clock, recieved clock) +1

All clocks start at 0 P3 starts at T=15 there (accumulated from prior ops)

Total messages = 10 (Tier 1 RA) + 12 (Tier 2 token) = 22 = M(72; 12)

Global token Request

Gateway broadcasts

G1 → G2 : Token ReQ (SN=101, T=21)

G1 → G3 : Token-ReQ (SN=101, T=21)

G1 → G12 : Token-ReQ (SN=101, T=21)

Token Transfer

Current token holder

G8 → G1 : Token (SN=101, T=22)

Critical Section Entry

G1 → P3 : Token Granted.

P3 → critical section

Exit –

node Exists CS.

Deffered Ricart-Agrawala replies are released.

Gateway retains token untill another cluster requests it.